Spinal muscular atrophy (SMA)
A blood test that now leads to treatment

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the doctor's.

Indian study (Kerala): Aswathy CG, Sankar VH, Alex ST, Santhi S, Dasaradh H. *Prevalence of exon 7/exon 8 deletion in patients with hypotonia and spinal muscular atrophy.* Indian Journal of Experimental Biology 2024;62(10):810-4. Genetic clinic of a tertiary care hospital in Kerala.

Who was tested

- 33 children with low muscle tone (floppy infants, weakness, tongue fasciculations)
- First-line test: MLPA for homozygous deletion of SMN1 exons 7 and 8
- Parents offered carrier testing

What was found

- **19 of 33 (58%)** had homozygous SMN1 deletion
- Type 1: 68.4%; type 2: 21%; type 3: 10.5%
- All parents tested were heterozygous carriers
- Authors: carrier testing is essential for prenatal diagnosis in later pregnancies

More Indian data

Carriers (North India, Kaur et al., J Family Med Prim Care 2022): of 198 adults with no family history, **1 in 30** carried an SMN1 deletion (MLPA and qPCR). **Test methods (AIIMS Jodhpur, Shekhawat et al., J Neurosci Rural Pract 2025):** ddPCR and MLPA gave identical SMN1/SMN2 copy results in 72 samples; ddPCR was cheaper and faster.

Facts (NINDS, MDA)

- About 1 in 10,000 births; loss of motor nerve cells in the spinal cord; thinking and learning are not affected. Autosomal recessive: both parents are usually healthy carriers.
- **Types:** 1 (before 6 months, never sits unaided); 2 (6-18 months, sits but cannot stand or walk); 3 (after 18 months, walks); 4 (adult onset).
- The SMN1 blood test identifies at least 95% of type 1, 2 and 3 cases. More SMN2 copies usually mean a milder course.
- **Treatments:** nusinersen (2016, intrathecal), onasemnogene abeparvovec (2019, one-time gene therapy under 2 years), risdiplam (2020, oral). Benefit is best documented when started early. A lower-cost generic risdiplam has been launched in India.
- Prenatal testing in a later pregnancy: disease only; fetal sex never disclosed (PCPNDT).

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SMA: how to approach doctors

Topic	What to do or say
Doctors to visit	Paediatric neurologists; paediatricians and neonatologists; physiotherapy and rehabilitation centres; orthopaedics (scoliosis) and pulmonologists (breathing).
Which children	Weak, floppy baby; not sitting or standing at the expected age; losing motor skills or falling more; tongue twitching; weak cry, feeding or breathing difficulty.
Opening question	"Doctor, in a weak, floppy baby, do you send the SMN1 test early?"
Facts to quote	Kerala: 58% of tested children with low tone had SMA. North India: 1 in 30 adults is a carrier. Treatment works best when started early.
Test to suggest	SMN1 deletion (copy number by MLPA or ddPCR), with SMN2 copy number; carrier testing for parents and siblings. **Not on our standard menu: confirm availability, MRP and report time with the lab first.**
"Let us do EMG first"	"The SMN1 blood test identifies at least 95% of cases (NINDS), without a needle test."
"Treatment is unaffordable"	"Risdiplam is taken by mouth, and a lower-cost generic has been launched in India. The choice is yours, doctor."
Sample	EDTA blood; parents for carrier testing; counselling and written consent first.
Never say	That treatment is a cure, or anything that blames the parents. Carriers are healthy.

References

- Aswathy CG et al. Prevalence of exon 7/exon 8 deletion in patients with hypotonia and SMA. Indian J Exp Biol 2024;62:810-4. or.niscpr.res.in
- Kaur A et al. Carrier frequency of SMA in individuals of reproductive age from North India. J Family Med Prim Care 2022;11:7870-4. doaj.org
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- NINDS. Spinal muscular atrophy. ninds.nih.gov · MDA. SMA fact sheet. mda.org
- MrMed. Risdiplam India launch. mrmed.in

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